XIAONAN LUO

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EDUCATION

University of Notre Dame *May 2028 (expected)*
Ph.D. in Computer Science and Engineering
Advisor: Xiangliang Zhang

The Hong Kong University of Science and Technology (HKUST) *July 2024*
B.S. in Data Science and Computer Science

RESEARCH INTERESTS

I study large language models (LLMs), with a focus on **autonomous research agents**, **security**, **evaluation**, **post-training**, **synthetic data**, and **reasoning**.

RESEARCH EXPERIENCE

Bake AI Aug 2026 – Oct 2026
Student Researcher

- Contributed to Épi, a self-evolving research framework whose agents preserve findings across sessions, explore new strategies, and verify advances against fixed evaluators; the project reported 120 record-setting mathematical results.

SELECTED PUBLICATIONS

SecProbe: Adaptive Evaluation of Coding Agents on Cybersecurity Vulnerabilities
X. Luo, Y. Huang, K. Guo, P. He, C. Zou, C. Gao, L. Li, Y. Ma, Z. Xu, Z. Chen, Y. Han, X. Zhang
arXiv:2609.33763, 2026

- Developed SecProbe, an adaptive evaluation framework that uses Item Response Theory to estimate coding-agent ability and select informative repository-scale vulnerability-repair tasks.
- Designed on-demand synthesis of executable vulnerability-repair tasks with exploit-based tests and reference patches when the existing task pool lacks suitable coverage.
- Built 353 tasks across six programming languages and 151 CWE types; evaluated nine frontier models with two agent harnesses (best success rate: 28.33%) and achieved comparable ability estimates with up to 29.5% fewer tasks than baselines.

Learning from Synthetic Data without Model Collapse in Iterative Instruction Tuning
X. Luo, Y. Huang, K. Guo, P. He, C. Zou, T. Hua, X. Zhang
Under review

- Identified model collapse in iterative instruction tuning as a polarization of competence, using DINA-based skill diagnosis to show that synthetic training can reinforce strong skills while degrading weakly supported ones.
- Proposed KITE, a two-stage synthetic-data framework that couples weakness-guided candidate generation with boundary-aware uncertainty curation, enabling more stable multi-generation improvement than strong synthetic-data baselines across reasoning benchmarks.

Better Datasets Start From RefineLab: Automatic Optimization for High-Quality Dataset Refinement
X. Luo, Y. Huang, P. He, X. Zhang
The 40th Annual AAAI Conference on Artificial Intelligence (AAAI), 2026

- Proposed RefineLab, a dynamic evaluation framework for LLMs that formulates QA dataset refinement as an integer linear program, automatically selecting coverage, difficulty, and factual-consistency edits under a token budget to align datasets with explicit quality targets.
- Designed and evaluated modular refinement operators, demonstrating stable evaluation improvements on benchmarks including MMLU, GSM8K, and HumanEval.

AdaReasoner: Adaptive Reasoning Enables More Flexible Thinking
X. Wang, Y. Huang, **X. Luo**, Y. Wang, K. Guo, Y. Zhou, X. Zhang
The Thirty-ninth Annual Conference on Neural Information Processing Systems (NeurIPS), 2025

- Developed AdaReasoner, an LLM-agnostic reinforcement learning plug-in that adaptively selects reasoning instruction format, temperature, and reasoning depth using a factorized action space, Boltzmann exploration, and a pretrained reward model.
- Demonstrated consistent gains over CoT, ToT, Auto-CoT, Best-of-N, and in-context CoT across six backbone LLMs, with strong few-shot efficiency, out-of-distribution robustness, and improved performance on knowledge-intensive tasks.

PUBLICATIONS INVOLVED

- Y. Huang, Z. Jiang, **X. Luo**, K. Guo, H. Zhuang, Y. Zhou, Z. Yuan, X. Sun, J. Schleinitz, Y. Wang, *et al.* "ChemOrch: Empowering LLMs with Chemical Intelligence via Groundbreaking Synthetic Instructions," *NeurIPS*, 2025. Link
- X. Wang, Y. Huang, Y. Zhou, **X. Luo**, K. Guo, X. Zhang. "Right Answers for the Right Reasons: Coherence-Enhanced Policy Optimization for Large Language Models," *EMNLP*, 2026. Link
- Y. Ma, Y. Huang, W. Wang, **X. Luo**, X. Zhang, S. Feuerriegel. "Synthetic Interaction Data for Scalable Personalization in Large Language Models," *KDD*, 2026. Link
- Y. Huang, Y. Jiang, W. Wang, H. Zhuang, **X. Luo**, Y. Ma, Z. Xu, Z. Chen, N. Moniz, Z. Lin, *et al.* "Emergent Social Intelligence Risks in Generative Multi-Agent Systems," *arXiv:2603.27771*, 2026. Link
- H. Zhuang, H. Yoo, **X. Luo**, K. Guo, X. Zhang. "Reliable Control-Point Selection for Steering Reasoning in Large Language Models," *arXiv:2604.02113*, 2026. Link

ACADEMIC SERVICE

Journal Reviewer: TKDD
Conference Reviewer: ICLR 2027, KDD 2026, COLM 2026

SKILLS

Programming Languages	Python, C/C++
Tools	PyTorch, verl, LLaMA-Factory, Transformers, Git, Docker